



GOVERNMENT POLYTECHNIC, AMRAVATI
(AN AUTONOMOUS INSTITUTE OF GOVT. OF MAHARASHTRA)
CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING/
INFORMATION TECHNOLOGY**

COURSE CODE: FC4452

COURSE TITLE: SOFTWARE PROJECT MANAGEMENT

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
IV-B		04	-		04	16	64	-	-

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	--	--	100
	MIN.	28	---	40	--	---	

@: Internal Assessment #: External Assessment

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1.RATIONALE:

Software Engineering is the foundation for professional processes to be followed involving principles, techniques, and practices for software development. The course provides a framework for software professionals for building quality assured software products. It enables students to blend the domain specific knowledge with the programming skills to create quality software products.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to:

1. Select suitable Software Process model for software development.
2. Prepare software requirement specifications.
3. Use Software modeling to create data designs.
4. Estimate size and cost of software product.
5. Use project scheduling techniques, tracking methods and reengineering principles in software development.
6. Apply software quality assurance in software development.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (in cognitive domain)	Topic and Sub-topics	CO No.	Hrs	Mks
Unit 1 Software Development Process	1a. Suggest the attributes that match with standards for the given software application 1b. Recommend the relevant software solution for the given problem with justification. 1c. Select the relevant software process model for the given problem statement with justification. 1d. Suggest the relevant activities in Agile Development process in the given situation with justification	1.1 Software, Software Engineering as layered approach and its characteristics, Types of software. 1.2 Software development framework. Software Process Framework. 1.3 Process models; Perspective Process Models, Specialized Process Models. 1.4 Agile Software development: Agile Process and its importance, Extreme Programming, Adaptive Software Development, Scrum, Dynamic Systems Development Method (DSDM), Crystal. Selection Criteria for software process model Waterfall, Incremental, RAID, Spiral.	1	10	12
Unit 2 Software Requirement Engineering	2a. Apply the principles of software engineering for the given problem. 2b. Choose the relevant 'requirement engineering' steps in the given problem. 2c. Represent the 'requirement engineering' model in the given problem. 2d. Prepare SRS for the given problem.	2.1 Software Engineering Practices and its importance, Core principles. 2.2 Communication Practices, Planning Practices, Modelling practices, construction practices, software deployment (Statement and meaning of each principle for each practice). 2.3 Requirement Engineering: Requirement Gathering and Analysis, Types of Requirements (Functional, Product, organizational, External Requirements),	2	10	12

		Eliciting Requirements, Developing Use cases, Building requirement models, Requirement Negotiation, Validation. 2.4 Software Requirement Specification: Need of SRS, Format, and its Characteristics.			
Unit 3 Software Modeling and Design	3a. Identify the elements of analysis model for the given software requirements. 3b. Apply the specified design feature for software requirements modeling. 3c. Represent the specified problem in the given design notation. 3d. Write the given characteristics of software testing. 3e. Prepare test cases for the given module.	3.1 Translating Requirement model into design model: Data Modeling. 3.2 Analysis Modeling: Elements of Analysis model. 3.3 Design Modeling: Fundamental Design Concept (Abstraction, Information hiding, Structure, Modularity, Concurrency, Verification, Aesthetics) 3.4 Design notation: Data Flow Diagram (DFD), Structured Flowcharts, and Decision Tables. 3.5 Testing- Meaning and purpose, testing methods- Black-box and White-box, Level of testing-Unit testing. 3.6 Test Documentation- Test Case Template, test plan, Introduction to defect report, test summary report.	3	12	12
Unit 4 Software Project Management	4a. Estimate the size of the software product using the given method. 4b. Estimate the cost of the software product using the given empirical method. 4c. Evaluate the size of the given software using COCOMO	4.1 The management spectrum-4P's 4.2 Metrics for Size Estimation: Line of Code (LoC), Function Points (FP). 4.3 Project Cost Estimation Approaches: Overview of Heuristic, Analytical, and Empirical Estimation. 4.4 COCOMO (Constructive	4	12	14

	model. 4d. Apply the RMMM strategy in Identified risks for the given software development problem.	Cost Model), COCOMO II. 4.5 Risk Management: Risk Identification, Risk Assessment, Risk Containment, RMMM Strategy.			
Unit 5 Project Scheduling & Reengineering	5a. Use the given scheduling technique for the identified project. 5b. Draw the activity network for the given task. 5c. Prepare the timeline chart/Gantt chart to track progress of the given project. 5d. State need of reengineering in software.	5.1 Project Scheduling: Basic principle Work breakdown structure, Activity network and critical path Method, Scheduling techniques (CPM, PERT). 5.2 Project Tracking: Timeline charts, Earned Value Analysis, Gant Charts 5.3 Reengineering Business Process Reengineering, Software Reengineering, Reverse Reengineering.	5	10	10
Unit 6 Software quality assurance and Security	6a. Write Software Quality Assurance (SQA) activity. 6b. Write features of the given software quality evaluation standard. 6c Use secure software engineering.	6.1 Software Quality Management Vs. Software Quality Assurance. Phases of Software Quality Assurance: Planning, Activities, audit, and review 6.2 Quality Evaluation standards: Six Sigma for software engineering, CMMI: Levels, Process areas, 6.3 McCall's quality factors. 6.4 ISO 9000 Quality standard. 6.5 Software Security, Introduction to DevOps, Secure software engineering.	6	10	10

4. LIST OF PRACTICALS: --NOT APPLICABLE

5. SUGGESTED STUDENT ACTIVITIES:

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a. Prepare journals based on practical performed in laboratory.
- b. Give seminar on relevant topics.
- c. Study and analyze college website from perspective of software application.
- d. Study and analyze any available application software from perspective of software engineering.

6. SUGGESTED INSTRUCTIONAL STRATEGIES:

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

- a. Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b. Use different types of teaching methods and media that are to be employed to develop the outcomes in addition to traditional lecture method.
- c. Ensure to create opportunities and provisions for **co-curricular activities**.
- d. Guide student(s) in undertaking micro-projects.
- e. Encourage students to refer different websites to have deeper understanding of the subject.
- f. About 10-15% of the topics/sub-topics/contents which is relatively simpler or descriptive in nature may be given to the students for **self-directed learning** and assess the development of the COs through classroom presentations. Keep the record of the topics/sub-topics/contents given to the students.

7. SUGGESTED MICRO-PROJECTS:

Only one micro-project is to be undertaken by a student in the beginning of the semester. She/he ought to submit it by the end of semester. Each micro-project should include two or more COs. The micro-project could be industry application based, internet based, laboratory based, field based and survey based. In all semesters, it could be group-based to build up skill and confidence in every student. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission.

The student ought to submit micro-project by the end of the semester to develop the industry oriented COs. A suggested list is given below. Similar micro projects could be added by concerned faculty.

- a. Study and analyze given software and write the characteristics and functions of the same.
- b. Case study of application specific software product for requirement engineering
 - i. Identify the problem statement
 - ii. Perform feasibility analysis
 - iii. Identify application specific requirements by following RE steps
 - iv. Prepare SRS
- c. Choose any problem statement and use data models to represent the solution.
- d. Prepare the test document of given software.
- e. Estimate the cost of given software.
- f. Survey on project risk management.

- g. Choose a problem, create activity network and use different project scheduling and tracking tools for the same.
- h. Case study of reengineering on given software project.
- i. Evaluate the quality of given software.
- j. Case study on software security.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

-----NOT APPLICABLE-----

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Software Development Process	12	4	4	4
2	Software Requirement Engineering	12	4	4	4
3	Software Modelling and Design	12	4	4	4
4	Software Project Management	14	2	6	6
5	Project Scheduling & Re-engineering.	10	2	4	4
6	Software quality assurance and Security	10	2	4	4
Total		70	18	26	26

R= Remember, U= Understanding, A=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of **UOs**. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

Sr.No.	Title Of Book	Author	Publication
1.	Software Engineering:A practitioner's approach	Pressman,Roger S.	McGrawHillHigher Education,NewDelhi ,(Seventh Edition)ISBN 978-0-07-337597-7
2.	Software Engineering Concepts	Fairly,Richard	McGraw Hill Education New Delhi-2001,ISBN-13:9780074631218
3.	Software Engineering:Principles and practices	Jain,Deepak	Oxford University Press,New Delhi ISBN 9780195694840

11. SOFTWARE/LEARNING WEBSITES:

- a. www.nptel.ac.in/courses/
- b. www.tutorialspoint.com//software_engg/
- c. www.sei.cmu.edu
- d. www.versionone.com/agile-101/

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	C P Ahir	Lecturer in Computer Engineering.	Govt. Polytechnic Amravati
2	K P Ukey	Lecturer in Information Technology	Govt. Polytechnic Amravati
3	S A Kale	Lecturer in Information Technology	Govt. Polytechnic Amravati

Govt. Polytechnic, Programme Board of Studies Computer Engineering has approved the above course curriculum on 20/12/18. and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 03/01/19

The Governing Body has approved the above course curriculum on 07/01/19